

Miguel Zumalacárregui

Group Leader, Astrophysical and Cosmological Relativity — Max Planck Institute for
Gravitational Physics (Albert Einstein Institute), Potsdam

Ramón y Cajal Fellow — Instituto de Física Teórica IFT-UAM/CSIC, Madrid

miguel.zumalacarregui@aei.mpg.de · migueltzuma.github.io · ORCID 0000-0002-9943-6490

The publication record is maintained on [INSPIRE-HEP](#); research [projects and funding](#) and the [people supervised](#) are listed on the website.

Positions

- **Ramón y Cajal Fellow**

Instituto de Física Teórica IFT-UAM/CSIC, Madrid · 2026–

- **Group Leader**, Astrophysical and Cosmological Relativity

Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam · 2020–

- **Marie Skłodowska-Curie Global Fellow**

Berkeley Center for Cosmological Physics, with return phase at IPhT Saclay · 2017–2020

- **Nordita Fellow**

Nordic Institute for Theoretical Physics, Stockholm · 2015–2017

- **Postdoctoral Researcher**

Institut für Theoretische Physik, University of Heidelberg · 2013–2015

- **Postdoctoral Researcher**

Instituto de Física Teórica IFT-UAM/CSIC, Madrid · 2013

Education

- **PhD in Physics**, *Cum Laude* with International Mention

Universidad Autónoma de Madrid · 2012 · Graduate work at ICC Barcelona and IFT-UAM/CSIC (FPI fellowship)

Dissertation: [Probing the Foundations of the Standard Cosmological Model](#)

- **Master's thesis**

Universidad Autónoma de Madrid · [PDF](#)

Scientific coordination

- **LISA Consortium** — member since 2018

Council member and Senior Co-Chair of the Cosmology Working Group (2025–); section coordinator for gravitational-wave lensing in the LISA Cosmology white paper (2022)

- **Euclid Consortium** — member since 2014

Lead of the Science Working Group on Gravitational Waves and of its “bright sirens” work package (2025–); member of the Strong Lensing, Cosmology Theory, and Supernovae & Transients working groups

- **GW-Space 2050**

Coordinator for the cosmography section, defining ESA’s gravitational-wave missions after LISA (2023–)

- **Multi-messenger lensing white paper**

Coordinator for the microlensing section (2024)

Workshops and meetings organized

- **13th LISA Cosmology Working Group Workshop**

Barcelona, 2026

- [12th LISA Cosmology Working Group Workshop](#)
Tallinn, 2025
- [Lensing and Wave Optics in Strong Gravity](#)
Erwin Schrödinger Institute, Vienna, 2024
- [LISA joint Fundamental Physics + Waveform meeting](#)
Potsdam, 2024
- [ACR Division Seminars](#)
Max Planck Institute for Gravitational Physics, 2020–2022
- [BCCP Cosmology Lunch](#)
UC Berkeley, 2016–2017
- [Cosmology Seminars, Institut für Theoretische Physik](#)
University of Heidelberg, 2014–2015
- [TRR33 Winter Schools](#)
Passo del Tonale, 2013, 2014 and 2015

Teaching

Graduate schools

- **Gravitational waves** — TAE (Taller de Altas Energías), Benasque, 2024
[Slides](#)
- **Tests of gravity and dark energy with cosmology and gravitational waves**
II Mesoamerican Workshop on Gravity and Cosmology, Chiapas, 2018
- **Alternative gravity theories** — Cosmology School in the Canary Islands, 2017
[Video lecture](#)
- **School for Cosmology Tools** — IFT Madrid, 2017
Lectures on CLASS, hi_class and MontePython
- **Workshop on CLASS and MontePython** — MPA Munich, 2014

University courses

- **Berkeley Connect Instructor** — UC Berkeley, 2018–2019
Two semesters of undergraduate mentoring and teaching
- **Lecturer** — University of Heidelberg, 2014–2015
Advanced Cosmology; Introduction to CMB Physics; Statistical Mechanics; advanced seminar
- **Teaching assistant** — University of Heidelberg, 2014–2015
Theoretical Astrophysics (graduate), with M. Bartelmann
- **Teaching assistant** — Universidad Autónoma de Madrid, 2013
Gravitation and Cosmology; Physics of the Cosmos
- **Mini-course on inhomogeneous cosmologies** — University of Barcelona, 2012

Awards and fellowships

- Ramón y Cajal Fellowship (2024).
- R3 certification, Spanish Research Agency.
- Berkeley Connect Fellow, UC Berkeley (2018).
- Marie Skłodowska-Curie Global Fellowship (2015).
- Nordita Fellowship (2015).
- Yggdrasil Fellowship (2011) — research stay at ITA Oslo.
- FPI doctoral fellowship (2008) and UAM Master’s scholarship (2007).

Evaluation and peer review

Evaluation panels

- NASA Hubble Fellowship Program, Physics and Cosmology panel.
- FCT — Portuguese Foundation for Science and Technology, Individual Support Call.
- Max Planck Institute for Gravitational Physics — postdoctoral hiring committee, since 2020.

External evaluator for funding agencies

European Research Council (Starting Grants), STFC/UKRI, Swiss National Science Foundation, Alexander von Humboldt Foundation, InterTalentum (Marie Skłodowska-Curie COFUND), Netherlands Research Council (NWO), Research Grants Council of Hong Kong, and the Estonian Research Council.

Refereeing

Peer review for more than 40 papers, including Physical Review Letters, Physical Review D, Nature Astronomy, JCAP, The Astrophysical Journal and Astrophysical Journal Letters.

Selected talks

- **Spanish & Portuguese Relativity Meeting** — Universidad de Murcia, 2026 · plenary
Gravitational lensing of waves: a new window into astrophysics, dark matter and gravity
- **IAS and Princeton University** — gravitational-wave meeting, Princeton, 2026 · seminar
GW231123 as a diffracted and magnified black-hole merger
- **Leibniz Institute for Astrophysics** — Potsdam, 2024 · institute colloquium
Gravitational lensing of waves
- **New Frontiers in Strong Gravity** — Benasque, 2024 · plenary
Wave-optical phenomena in gravitational-wave lensing
- **Workshop on Tensions in Cosmology** — Corfu, 2022 · plenary
Towards solutions to the Hubble problem beyond Einstein's gravity
- **EuCAPT Annual Symposium** — CERN, 2022 · plenary review
[Dark energy and tests of gravity with cosmology and gravitational-wave propagation](#)
- **Institute of Theoretical Astrophysics** — University of Oslo, 2021 · institute colloquium
The dark Universe in the era of gravitational-wave astronomy
- **Progress in Old and New Themes in Cosmology** — Palais des Papes, Avignon, 2020 · plenary
Tests of gravity with gravitational waves
- **COSMO** — Aachen University, 2019 · plenary
Tests of dark energy and modified gravity
- **Rencontres du Vietnam** — Very High Energy Phenomena in the Universe, 2018 · plenary
The dark Universe in the era of gravitational-wave astronomy

Languages

Spanish (native), English (C2), German (C1).